

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

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STUDENT-TEACHER-SUBJECT CODE DATABASE

Aim:

To implement a Prolog database representing the relationship between students, teachers and subjects.

Objective:

To store student information.

To associate teachers with subjects.

To determine which teacher teaches a particular subject. To determine which subjects are taken by a student.

Algorithm: Start.

Define student-subject facts. Define teacher-subject facts.

Create rules to relate students, subjects and teachers. Query the required relationship.

Display the result. Stop.

Flowchart:

START

| v

Store Student-Subject Facts

| v

Store Teacher-Subject Facts

| v

Enter Query

| v

Match Facts/Rules

| v

Display Result

| v

STOP

Prolog Program:

% Student-Teacher-Subject Database

student(rahul, ai). student(rahul, mathematics). student(priya, ai). student(priya, physics). student(arun, mathematics). student(arun, physics).

teacher(ramesh, ai).

teacher(suresh, mathematics). teacher(kavitha, physics).

teaches(Teacher, Student, Subject) :-student(Student, Subject),

teacher(Teacher, Subject).

Output:

Result:

Thus, the Prolog program successfully represents and queries student-teacher-subject relationships.